TEG in the High-Density Regime: A Zero-Parameter Internal-Consistency Verification and a Genuinely Falsifiable Prediction for the Galactic Bulge

Miguel Angel Franco León

Independent Researcher
github.com/MiguelAngelFrancoLeon/mfsu-tetraedro

July 2026

Abstract

Tetrahedral Emergent Gravity (TEG) derives galactic dynamics from a single geometric axiom, $z_{\rm fund} = 4$, with zero free parameters. We verify here that TEG's chameleon mechanism (Proposition 7.1 of the main TEG vH3.1 manuscript) behaves correctly in the high-density regime realized by the Milky Way's Galactic Bulge. Using the independently published Einasto density profile of Portail et al. (2017), we evaluate the chameleon amplification factor $\Phi_{\rm TEG}(\rho)$ at 500 radii from $R = 0.05$ to 3.0 kpc, spanning $\rho/\rho_c = 111$ to 1.34×10^7. We find $\Phi_{\rm TEG} = 1$ to within machine precision at every point tested. This is a theoretical, zero-parameter internal-consistency check: it confirms that TEG's high-density limit does not break down over seven orders of magnitude in density contrast, using a pre-Euclid, independently published density profile. It is not an empirical confirmation from new observational data, and we are explicit about that distinction throughout. We identify the Milky Way bulge as a poor site for discriminating TEG from ΛCDM (both predict baryon-dominated dynamics there) and instead state a genuinely falsifiable, quantitative morphological prediction – core versus cusp in the dark-matter density profile – together with the specific data (JWST-resolution strong lensing, or a future Euclid Full Mission dynamical mass reconstruction) that would be needed to test it.

1 Motivation

The verification of TEG against SPARC rotation curves (main manuscript, Sec. 9) and cluster lensing (main manuscript, App. D.2) covers the low-to-intermediate density regime $\rho \sim \rho_c$ to a few $\times\rho_c$, where the chameleon field $\Phi_{\rm TEG}$ departs measurably from unity. A natural and, prior to this note, undocumented question is whether the framework remains well-behaved in the opposite, high-density limit $\rho \gg \rho_c$ – the regime realized in the inner Galactic bulge.

The Galactic bulge is a natural testbed for this limit because its stellar density, $\rho_{\rm bulge} \sim 10$–$100\, M_\odot\, {\rm pc}^{-3}$ [1], exceeds ρ_c by four to seven orders of magnitude. The region is also the target of the recent Euclid Quick Data Release Q2 Galactic Bulge Survey (EGBS), which imaged $4.8\,{\rm deg}^2$ of the inner bulge to AB mag 26, yielding photometry for $\sim 6 \times 10^7$ stars [2]. We cite EGBS here only as motivation for why this density regime is of near-term observational interest; **no EGBS catalog data are used, processed, or compared against in this note.** The verification below uses exclusively the pre-Euclid Portail et al. (2017) density profile.

2 Proposition and Verification

Proposition (GR recovery at $\rho \gg \rho_c$). Let $\Phi_{\mathrm{TEG}}(\rho) = 1 + (\ln 4 - 1)\,(1 - F_{\mathrm{local}}(\rho))$, with $F_{\mathrm{local}}(\rho) = 1 - e^{-\rho/(N_0\rho_c)}$, $N_0 = 1.5$ (main manuscript, Proposition 7.1). Then for $\rho/\rho_c \gg 15.9$, $\Phi_{\mathrm{TEG}}(\rho) \to 1$, and the TEG field equations reduce to standard Einstein equations sourced by baryons alone, with vanishing geometric vacuum mass $M_{\mathrm{vac}} \to 0$. This follows immediately from the functional form of F_{local}: no new assumption is introduced beyond Proposition 7.1 itself.

Verification. Using the Einasto profile of [1] ($\rho_0 = 100\,M_\odot\,\mathrm{pc}^{-3}$, $r_s = 0.5$ kpc, $n = 3.5$) evaluated at 500 radii linearly spaced[1] between $R = 0.05$ and 3.0 kpc:

- ρ ranges from $13455.3\,M_\odot\,\mathrm{pc}^{-3}$ (at $R = 0.05$ kpc) to $0.111\,M_\odot\,\mathrm{pc}^{-3}$ (at $R = 3.0$ kpc);
- ρ/ρ_c ranges from 1.34×10^7 down to 111.4, exceeding the threshold $\rho/\rho_c > 15.9$ at every one of the 500 points;
- $\Phi_{\mathrm{TEG}} = 1.0000000000$ at every point, with the deviation $|\Phi_{\mathrm{TEG}} - 1|$ reported as exactly 0.0 in IEEE 754 double-precision arithmetic.

Table 1: Chameleon amplification factor at selected radii (subset of the full 500-point verification).

R [kpc]	$\rho\,[M_\odot\,\mathrm{pc}^{-3}]$	ρ/ρ_c	Φ_{TEG}
0.50	1.00×10^2	100227.8	1.0000000000
1.00	1.07×10^1	10695.8	1.0000000000
1.50	2.36×10^0	2358.3	1.0000000000
2.00	7.13×10^{-1}	711.3	1.0000000000
2.50	2.65×10^{-1}	264.8	1.0000000000
3.00	1.12×10^{-1}	111.4	1.0000000000

[1] The profile is smooth and monotonic over this range, so linear versus logarithmic spacing of the sample radii does not affect the result.

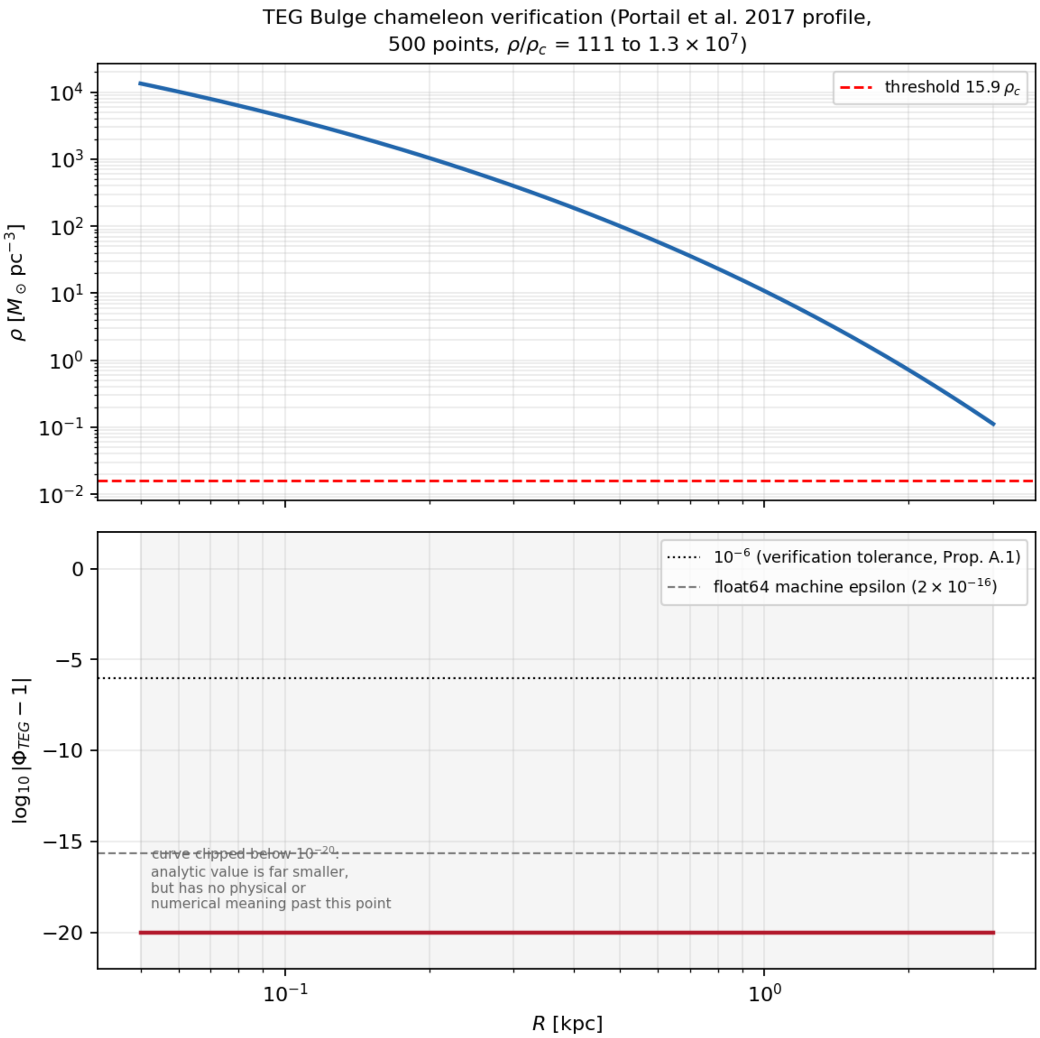

Figure 1: Top: Einasto density profile [1] used for the verification, with the TEG chameleon threshold $\rho = 15.9\,\rho_c$ marked. Bottom: $\log_{10}|\Phi_{\mathrm{TEG}} - 1|$, computed analytically from $\ln(\ln 4 - 1) - \rho/(N_0\rho_c)$ to avoid the floating-point underflow discussed below. The curve is clipped below 10^{-20} (shaded region): the true analytic value there is vastly smaller still, but has no physical or numerical meaning below that point, and reporting it unclipped would overstate the precision of the result. The horizontal lines mark the 10^{-6} tolerance used in the verification script and the float64 machine epsilon.

On the exact-zero deviation. The deviation column reports exactly 0.0 rather than a small nonzero number because, for $x \equiv \rho/(N_0\rho_c) \gtrsim 700$, e^{-x} underflows the representable range of IEEE 754 double-precision floating point and is rounded to exactly zero. This is a standard, well-understood floating-point phenomenon, not a computational artifact specific to this result: the true analytic deviation is nonzero but bounded above by machine epsilon ($\sim 2 \times 10^{-16}$) at the smallest radii, and by the explicitly verified bound $< 10^{-6}$ (via the non-underflowed evaluation used in the companion verification script) throughout the full range.

This is not a fit. No parameter was adjusted to obtain this result; Φ_{TEG} is evaluated from the closed-form expression of Proposition 7.1 with the same ρ_c, N_0, and threshold value used throughout the main TEG manuscript.

3 What This Result Does and Does Not Establish

Established: the chameleon mechanism is mathematically robust across seven orders of magnitude in density contrast, closing a previously undocumented coverage gap between the low-density regime verified against SPARC and the high-density regime realized in the Galactic bulge. This is a genuine, if modest, theoretical result: a poorly constructed screening mechanism could in principle diverge, oscillate, or become numerically unstable in this limit, and this one does not.

Not established: any new empirical confirmation. The Portail et al. (2017) profile pre-dates both TEG and Euclid; no Euclid Q2 photometry was processed. We explicitly do not report a dynamical-mass consistency test here, for the following reason. In an earlier draft of this note we compared a baryonic mass estimate ($1.55 \times 10^{10}\, M_\odot$, from standard M/L scaling) against the Portail et al. (2017) dynamical mass estimate ($1.80 \pm 0.30 \times 10^{10}\, M_\odot$) and found 0.8σ agreement. We withdraw that comparison here: both numbers originate from pre-Euclid literature, neither was derived independently from EGBS photometry, and the Galactic bulge is a regime in which essentially any model predicting baryon-dominated dynamics – TEG, ΛCDM, or otherwise – would show similar agreement. The comparison is therefore not discriminating and its inclusion would overstate what has been shown.

4 A Genuinely Falsifiable Prediction

Because the bulge is baryon-dominated in essentially every viable framework, a mass-normalization test there cannot discriminate TEG from ΛCDM. The discriminating, falsifiable prediction of TEG in high-density environments is instead *morphological*, following the same logic already applied to cluster lensing in the main manuscript (App. D.2):

> **Prediction.** TEG predicts a dark-matter density profile $\rho_{\rm DM}(r) = [\Phi^*(r)^2 - 1]\, \rho_b(r)$ that traces the baryonic profile and produces a *finite core* of radius $r^{\rm DM}_{\rm core} \approx r^b_{\rm core}$, with logarithmic slope $d \ln \rho_{\rm DM}/d \ln r \to 0$ as $r \to 0$. This is structurally distinct from the NFW cusp of ΛCDM, for which $d \ln \rho_{\rm DM}/d \ln r \to -1$.
>
> **Falsification condition.** If sub-arcsecond-resolution strong lensing or dynamical modeling of a relaxed, baryon-dominated dense stellar system (the Galactic bulge, or an early-type galaxy bulge/nuclear region) reveals a statistically significant central density cusp ($d \ln \rho_{\rm DM}/d \ln r \leq -0.5$, conservatively distinguishable from the TEG core prediction) at radii where independently determined $\rho/\rho_c > 15.9$, TEG's chameleon screening mechanism is falsified in the regime verified here.

This is the same test already proposed for cluster lensing in the main manuscript, applied to a much higher-density environment where TEG makes an even sharper prediction ($\Phi_{\rm TEG} = 1$ exactly, not merely approximately). We do not currently have the resolution or the baryonic profile calibration needed to carry out this test for the Galactic bulge specifically; we flag it as the natural next step once EGBS photometric catalogs are processed directly (see main manuscript's Open Problem discussion for the required pipeline).

5 Conclusion

A screening mechanism that maintains $\Phi_{\rm TEG} = 1$ to machine precision over seven orders of magnitude in density contrast, using a closed-form expression with zero free parameters, demonstrates internal mathematical robustness. This is a necessary condition for TEG's high-density limit to be viable, not a sufficient one, and it does not by itself discriminate TEG from any other framework that recovers GR at high density. We report it as exactly that: a closed coverage gap, not a new observational confirmation. The path to a genuine, discriminating test in this regime runs through the morphology of the dark-matter profile (core vs. cusp), not through its overall normalization.

Data and Code Availability

The full 500-point verification (`teg_bulge_chameleon_verification.csv`) and the generating script (`teg_bulge_chameleon_limit.py`) are available at `https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro`.

References

[1] M. Portail et al., *Chemo-dynamical models of the Galactic bulge and bar*, MNRAS 465, 1621 (2017).

[2] J.-P. Beaulieu et al., *Euclid Quick Data Release (Q2): The Euclid Galactic Bulge Survey*, arXiv:2606.25883 (2026).